Nuclear Physics

01.05 · Original practice problems

Show your reasoning, use SI units where appropriate, and state any assumptions. Constants are supplied in the questions that need them.

CONTENTS

01 Mass defect and binding

02 Radioactive clock

03 Fusion and energy

04 Neutrinos and stars

A MAIS Physics & Astronomy Club study resource. These are original exercises, not official IOAA problems.

PROBLEMS

01 Mass defect and binding

A nucleus contains two protons and two neutrons. The separate nucleons have total mass 4.03188 u and the nucleus has mass 4.00151 u. Use 1 u c^2 = 931.5 MeV. (a) Find the mass defect in u. (b) Find the total binding energy and binding energy per nucleon in MeV. (c) Explain why mass is lower for the bound nucleus than for its separated components.

02 Radioactive clock

A sample initially contains 8.0×10^{18} radioactive nuclei with a half-life of 5.0 days. (a) How many remain after 15 days? (b) Find the decay constant in s^{-1} using $\lambda = \ln 2 / t_{½}$. (c) Find the initial activity in becquerels using $A = \lambda N$. Assume the half-life is constant.

03 Fusion and energy

A simplified fusion reaction converts four protons into a helium nucleus and other products, releasing 26.7 MeV per reaction. Use 1 eV = 1.60×10^{-19} J and proton mass 1.67×10^{-27} kg. (a) Find the energy released per reaction in joules. (b) Estimate the power produced if 1.0×10^{11} kg of protons are consumed per second. (c) Explain why this estimate need not equal the energy carried away by photons alone.

04 Neutrinos and stars

In the solar interior, nuclear reactions produce neutrinos as well as photons. (a) Why can neutrinos escape from the Sun much more readily than photons? (b) How does detecting solar neutrinos give information about the solar core? (c) Give one reason why detecting these particles is difficult. Keep the discussion qualitative.